Geophys. J. R. astr. Soc. (1976) **46**, 295–305.

Palaeomagnetism of the Lower Cretaceous Sierra De Los Condores Group Cordoba Province Argentina

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(Received 1976 February 24; in original form 1975 August 18)

Summary

Measurements are described of the directions of remanent magnetization of 89 samples from nine lava flows and red beds. Stable remanent magnetization was isolated after AC demagnetizing. All the units have normal remanent magnetization, except one lava flow which yields a direction toward the north with positive inclination. From the mean direction of stable remanence, referred to the bedding, of each unit a virtual geomagnetic pole is computed; the mean of eight of these poles is 90·6 °E, 84·2° South, $\alpha_{95} = 4{\cdot}7°$ and represents the position of the palaeomagnetic pole for the exposures of the Sierra de Los Condores group from El Estrecho–Cerro Libertad. The position of this pole is reasonably close to the positions of the South American Lower Cretaceous palaeomagnetic poles for the Serra Geral and Vulcanitas Cerro Colorado formations and the trachybasaltic dykes from Rio Los Molinos. This supports the interpretations that the South Atlantic Ocean was formed in Lower Cretaceous times and that the Earth's magnetic field was on average similar to that of a geocentric dipole in South America in the Lower Cretaceous, and suggests that there has not been substantial relative movements between Central Argentina and Southern Brazil.

1. Introduction

Late Palaeozoic and Early Mesozoic palaeomagnetic poles for South America and Africa overlap when these continents are joined by their Atlantic margins (Valencio & Vilas 1972). In this reconstruction the only Lower Cretaceous South American palaeomagnetic pole available—Serra Geral formation, Creer (1964)—does not superpose with its contemporaneous African palaeomagnetic pole, suggesting a pre-Lower Cretaceous age for the formation of the South Atlantic Ocean. Valencio (1972) indicated that more Cretaceous palaeomagnetic poles for South America are necessary to sustain this interpretation.

Lower Cretaceous igneous and sedimentary rocks crop out at Sierra de Los Cóndores, in Cordoba Province, Argentina. Gordillo & Lencinas (1967) have included these rocks in the Sierra de Los Cóndores Group. Valencio (1972) reported the results of the palaeomagnetic study of the igneous and sedimentary rocks exposed

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at Cerro Colorado and included in the Vulcanitas Cerro Colorado formation of the Sierra de Los Condores group. In this paper results are reported of the palaeomagnetic study of the igneous and sedimentary rocks exposed at El Estrecho—Cerro Libertad and included in the Vulcanitas Cerro Colorado formation and in the overlying Vulcanitas Rumipalla formation of that group.

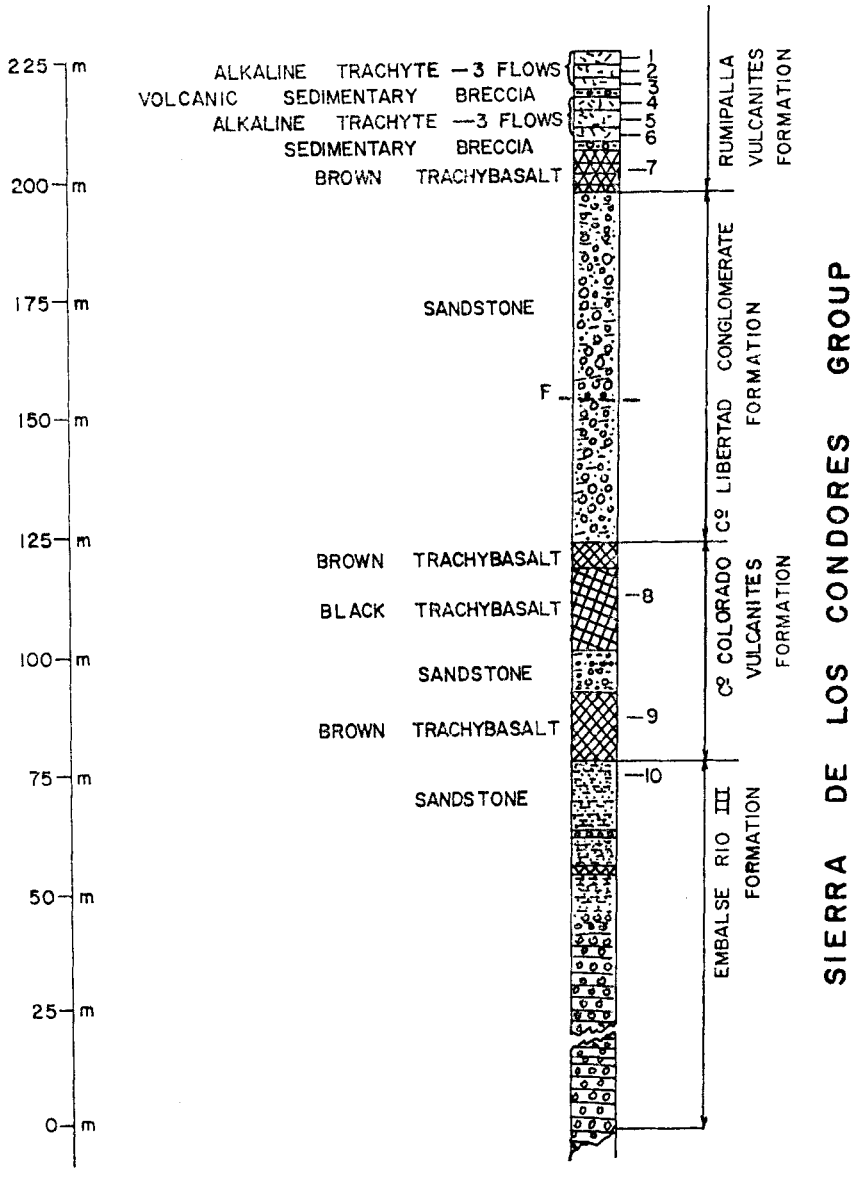


FIG. 1. Sierra de Los Condores group. Schematic stratigraphic column at El Estrecho-Cerro Libertad (Cordoba Province Argentina) according to Gordillo & Lencinas (1967). The units included in the palaeomagnetic study are indicated by numbers.

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2. Geological setting

The Sierra de Los Condores is located at the Southern part of Cordoba province (Argentina) and belongs to the Sierras Pampeanas system. In this area the metamorphic plutonic basement is overlain unconformably by a sedimentary-volcanic group about 200 m thick composed of continental detrital rocks which alternate which vulcanics of the alkaline–olivine basaltic association. Gordillo & Lencinas (1967) proposed the name Sierra de Los Condores for this group and assigned an albian–aptian age to it. The Sierra de Los Condores group consists of four formations which in chronological order from the base to the top are: Embalse Rio Tercero formation, Vulcanitas Cerro Colorado formation, Conglomerado Cerro Libertad formation, and Vulcanitas Rumipalla formation (Fig. 1).

The rocks of the Sierra de Los Condores group were sampled at about 64·1° longitude West and 32·2° latitude South (El Estrecho–Cerro Libertad); Fig. 1 shows a schematic stratigraphic column for this area (Gordillo & Lencinas 1967). The deposits included in the Embalse Rio Tercero formation, about 80 m thick, are mainly red beds; their lower part consists of sandy conglomerates and their upper part of sandstones and layered lime–sandstones with interbedded evaporites and tuffites. The sedimentary process which gave origin to the Embalse Rio Tercero formation was stopped by a volcanic cycle whose lavas (seven or more) were extruded conformably with the underlying sedimentary strata. Gordillo & Lencinas (1967) proposed the name of Vulcanitas Cerro Colorado formation for this vulcanics and sedimentary association. Only three lava flows are exposed in the sampling area (Fig. 1). The first of these lava flows is a brown trachybasalt which overlies the sediments of the Embalse Rio Tercero formation (bedding plane: strike = 355° dip = 11°) and shows a sharp contact with them; the baked zone in this sedimentary unit does not exceed 30 cm.

The Vulcanitas Cerro Colorado formation is overlain by a polymict orthoconglomerate, about 60–80 m thick, whose composition is similar to the conglomerate present in the base of the Embalse Rio Tercero formation. Gordillo & Lencinas have proposed the name of Conglomerado Cerro Libertad formation for this orthoconglomerate. A new volcanic cycle interrupted the sedimentary process. Seven lava flows extruded during this volcanic cycle are exposed in the sampling area (Fig. 1); they have been included in the Vulcanitas Rumipalla formation (Gordillo & Lencinas 1967). The oldest of these lava flows is a brown trachybasalt which overlies conformably the conglomerates of the Conglomerado Cerro Libertad formation (bedding plane: strike = 66° dip = 7°).

3. The palaeomagnetic study

Eighty-nine hand samples were collected from the Embalse Rio Tercero, Vulcanitas Cerro Colorado and Vulcanitas Rumipalla formations (Table 1). All hand samples were oriented by Sun and magnetic compass. In many cases magnetic azimuths of the samples exceed in 10° or more Sun–compass azimuths, For this reason the Sun–compass azimuths have been used for the field correction.

Six or more samples were collected at different sites, units1–4 and 5–9 in order to reduce the effects which might have been produced by spurious local magnetizations (Table 1). Five cylinders 2·5 cm in diameter and one cm in length were cut from each hand sample. Spinner and flux-gate magnetometers were used to measure the natural remanent magnetization of these cylinders. One cylinder from each sample was subjected to AC demagnetizing fields increasing in steps of 50 Oe up to 300 Oe peak field. In this manner the best AC demagnetizing field for isolating the stable remanence of each sample was determined. The criterion used for the choice of the best demagnetizing field strength was that the direction of the remanence of the pilot cylinder showed no further change in response to higher demagnetizing field strengths.

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Table 1

Summary of the palaeomagnetic data for the units of the Sierra de Los Condores group from El Estrecho–Cerro Libertad, Cordoba Province, Argentina.

Formation		Unit	Number of Samples or units collected	Number of Samples or units useful N	Mean direction of NRM (referred to the palaeohorizontal) D	I	k	α_{95}	Mean direction of cleaned remanent magnetization (referred to palaeohorizontal) D	I	k	α_{95}	Name	Virtual geomagnetic pole Lat. (°) South	Long. (°) East	$d\psi$ (°)	$d\chi$ (°)
	1	Vesicular Alkaline Trachyte	14	12	53·8	−41·3	1·8	47·2	0·6	−39·5	28·8	8·2	1	80·2	119·5	5·9	9·9
	2	Massive Alkaline Trachyte	21	21	9·9	−42·0	9·4	10·9	4·6	−51·1	57·2	4·2	2	86·1	200·5	3·9	5·7
Vulcanitas	3	Vesicular Alkaline Trachyte	6	6	334·8	−38·7	1·5	87·2	350·1	−41·5	101·4	6·6	3	77·9	66·9	5·0	8·2
	4	Massive Alkaline Trachyte	12	9	353·7	−50·5	28·8	9·7	349·7	−44·3	82·4	5·7	4	79·1	57·7	4·5	7·2
Rumipalla	5	Vesicular Alkaline Trachyte	2	2	3·6	−49·4	96·3	25·7	355·2	−45·5	138·4	21·4	5	83·4	76·2	17·3	27·2
	6	Vesicular Alkaline Trachyte	2	2	40·8	77·7	253·6	15·7	36·6	75·8	221·2	16·8	6	9·8	311·7	28·5	30·9
	7	Brown Trachybasalt	11	11	4·4	−40·7	4·1	25·4	2·5	−35·9	50·4	6·5	7	77·5	126·9	4·4	7·5
Vulcanitas	8	Black Trachybasalt	6	6	306·0	−50·4	2·3	57·1	359·3	−53·3	95·7	6·9	8	88·2	313·6	6·6	9·6
Cerro Colorado	9	Brown Trachybasalt and baked sediments	14	14	345·9	−48·9	16·0	10·2	356·0	−48·9	100·8	4·0	9	85·8	59·1	3·5	5·2
Embalse Rio III	10	Red beds	1	1	337·5	−48·8	—	—	322·3	−56·4	—	—	10	58·8	6·4	—	—
—		—	10	8	357·1	−48·9	14·3	15·1	357·1	−45·1	130·1	4·8		83·8	90·7	3·8	6·1

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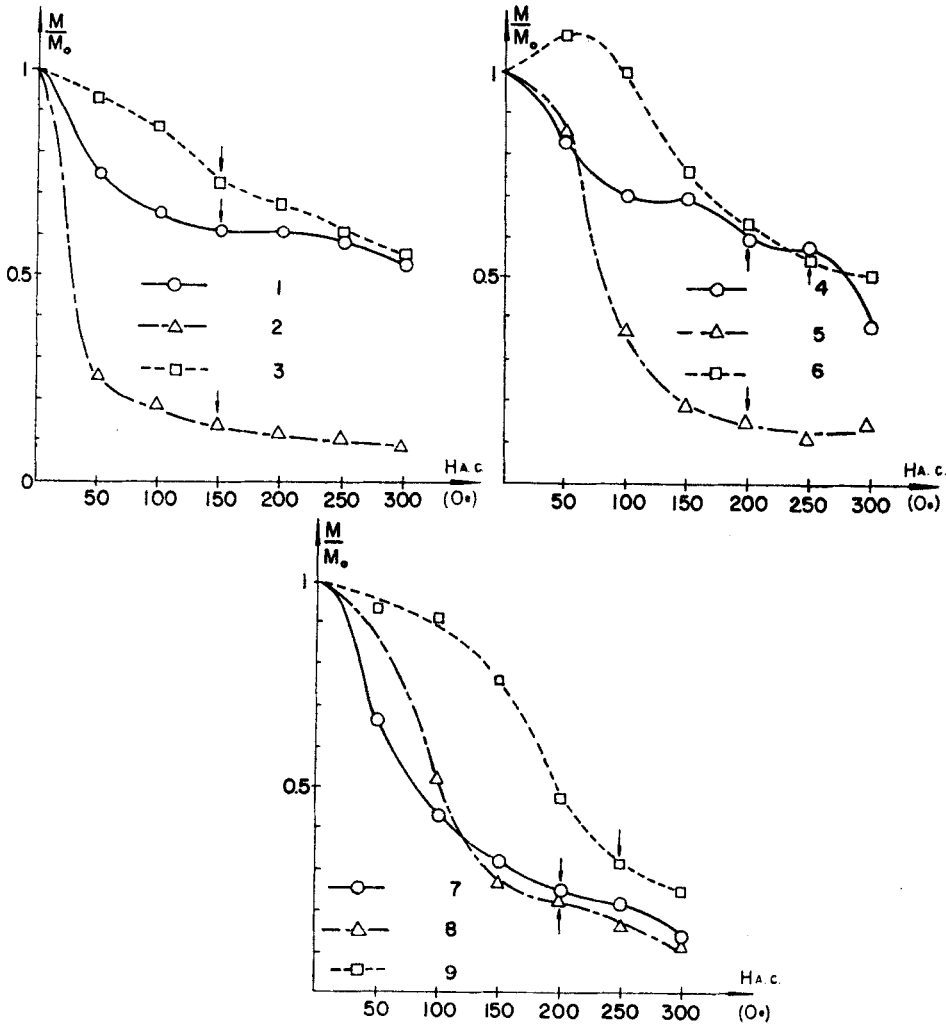


FIG. 2. Relative strength of the remanent magnetization after the different AC demagnetizing steps for typical samples from each geological unit from the Sierra de Los Condores group (the arrows indicate the best demagnetizing field strength for each sample).

Fig. 2 shows the relative strength of the remanent magnetization after the different AC demagnetizing steps for a typical sample from each geological unit. The best demagnetizing field strength for each typical sample is shown by an arrow in Fig. 2; it can be seen that it varies between 150 and 250 Oe. A second and sometimes a third cylinder of the sample were submitted to this best demagnetizing field for a better definition of the stable remanence of the sample. The direction of the stable remanence of the baked sediments is coincident with those of the basalts and give additional support to the stability of the remanence isolated in the brown trachybasalt of the Vulcanitas Cerro Colorado formation.

The statistical method of Fisher (1953) was used to compute the mean direction of the NRM and stable remanence of each unit on the basis of the NRM and stable remanence of its samples, respectively, (Table 1). It can be seen that the within-site

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precision is good for the units 2, 4, 5, 6 and 9, with k values as high as 254 for unit 6. On the other hand, the within site precision is bad for the other sites, with k values as low as 1·5 and 1·8 for sites 3 and 1, respectively. Directions of the stable remanence of samples from each of the units are well grouped; all the units are better grouped after AC demagnetization than before, except unit 6 which practically maintains its original clustering. This indicates that the AC cleaning has been suitable to eliminate secondary remanent magnetizations. Note that unit 6 yields a direction toward the north with positive inclination; it suggest that this unit may have been extruded during a rapid reversal of the geomagnetic field; all the other units have normal stable remanence.

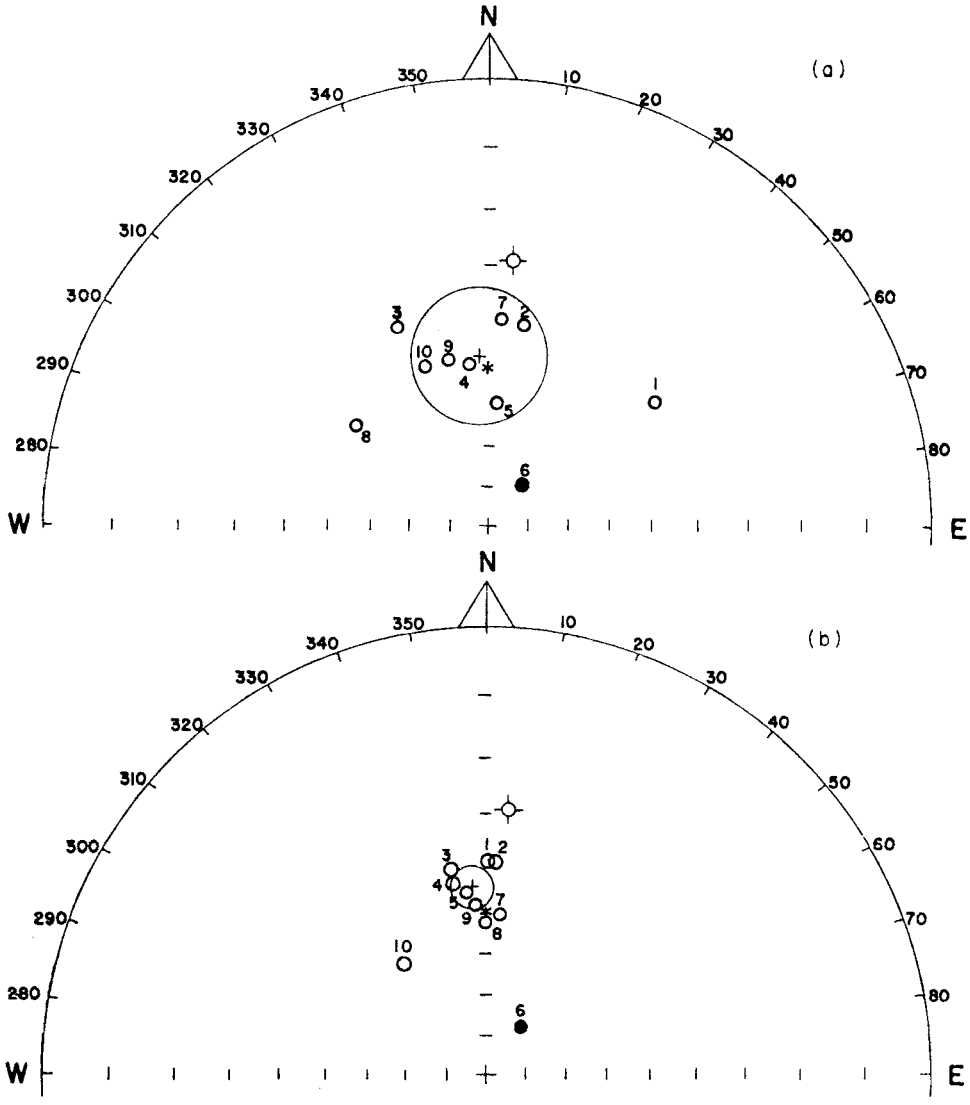


FIG. 3(a). Mean directions of the NRM, and (b) stable remanent magnetization, as referred to the palaeohorizontal, for each unit of Sierra de Los Condores group (Table 1). The mean directions (+) of these remanences (Table 1) are also shown. ⊹ Present geomagnetic field. * Axial dipole field.

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Table 2

Lower Cretaceous palaeomagnetic poles for South America.

		Age	Number of PGVs or PGs	Polarity	Palaeomagnetic pole				Name	Ref.
					Long. (°) East	Lat. (°) South	k	α_{95} (°)		
1	Serra Geral formation, Brazil	115–125 My	30	*N–R*	54·1	78·2	22·5	5·7	SAK_1	(1)
2	Vulcanitas Cerro Colorado formation, Arg.	118 ± 6 My	6	*R*	14·0	81·0	25·7	13·4	SAK_2	(2)
3	Dikes from Rio de Los Molinos, Arg.	129–150 My	4	*N–R*	8·4	78·8	26·9	18·0	SAK_3	(3)
4	Sierra de Los Cóndores group, Arg.	Albian–Aptian	8	*N*	90·6	84·2	139·8	4·7	SAK_4	
5	Mean of SAK_2, SAK_3, AAK_4	K(1)	3	*N–R*	26·0	82·5	166·3	9·5	SAK_{2-4}	
6	Mean of SAK_1, SAK_2, SAK_3, SAK_4	K(1)	4	*N–R*	36·1	81·7	178·0	6·9	SAK	

References: (1) Creer 1964; (2) Valencio 1972; (3) Linares & Valencio 1975.

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4. Interpretation of results

The normal cleaned remanent magnetization found in most of the units of the Sierra de Los Condores group from El Estrecho–Cerro Libertad contrasts with the reversed stable remanence isolated in the units of this Group collected at Cerro Colorado (Valencio 1972).

Mean directions of the remanent magnetization for each unit from El Estrecho–Cerro Libertad as referred to the palaeohorizontal, before and after AC demagnetization, are shown in Figs 3(a) and 3(b), respectively. As can be seen in Fig. 3 (b) the mean direction of the stable remanence of unit 6 is conspicuously outside the group formed by the directions of the stable remanence of the remaining units. If we postulate that unit 6 was extruded during a reversal of the geomagnetic field (Section 3) we must not include it in the computation of the mean direction of the stable remanent magnetization for the Sierra de Los Condores group at the locality El Estrecho–Cerro

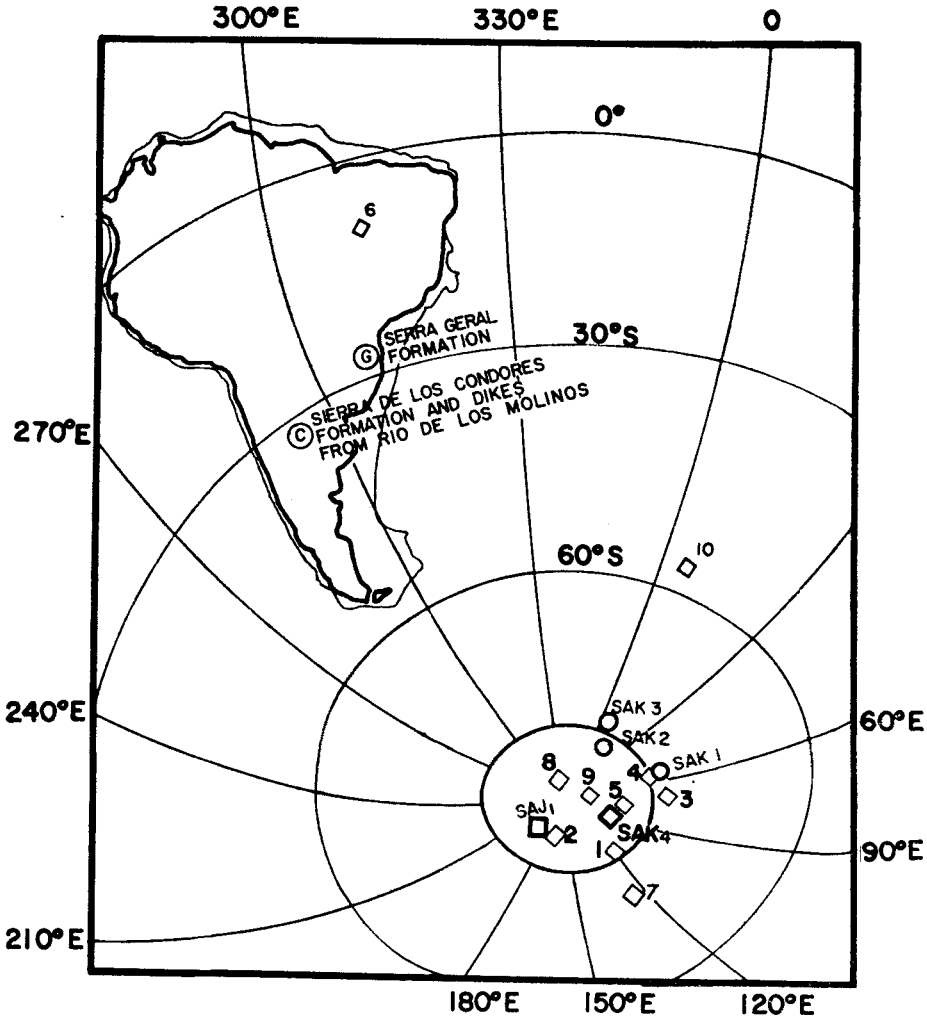


FIG. 4. Virtual geomagnetic poles (◇) for 10 units of Sierra de Los Condores group (Table 1) and Lower Cretaceous (○) and Middle Jurassic (□) palaeomagnetic poles for South America (Table 2). Other references in Section 4.

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Libertad. On the other hand the direction of the stable remanence for unit 10 has been determined on the basis of one sample; therefore we must not give it the same weight as the other units in the computation of the mean magnetization of the group. Thus the mean directions of the NRM and stable remanent magnetization for the Sierra de Los Condores group at the locality of El Estrecho–Cerro Libertad must be computed on the basis of the remaining eight geological units. These mean directions are given in Table 1 and shown in Figs 3(a) and 3(b). It can be seen that the directions of the stable remanences of units are substantially better grouped after AC demagnetization ($k = 130$) than before ($k = 14$). This supports the interpretation that substantial secondary remanences were destroyed during the AC demagnetization (Section 3).

The palaeomagnetic pole for the eight useful units of the Cerro de Los Condores group from El Estrecho–Cerro Libertad may be computed as the mean of their virtual geomagnetic poles (SAK_4 Table 2) or from the mean direction of their stable remanences (Table 1). Fig. 4 shows the position of SAK_1. Table 2 summarizes the positions of the palaeomagnetic poles of Cretaceous age computed for South America; the positions of these palaeomagnetic poles are shown in Fig. 4. In this figure are also indicated the areas where the rocks from the States of Rio Grande do Sul, Parana and Sao Paulo, Brazil (G, palaeomagnetic pole SAK_1) and from the Province of Cordoba, Argentina (C, palaeomagnetic poles SAK_2, SAK_3 and SAK_4), with which these poles have been computed, were sampled; the distance between these two areas is about 15°. Table 2 shows the mean of the positions of the palaeomagnetic poles computed for the rocks from the Province of Cordoba (entry 5). This mean polar position is statistically coincident with the position of the pole for the Serra Geral formation, Brazil. That suggests that on average the geomagnetic field in South America was similar to that of a geocentric dipole in Lower Cretaceous times and that the South American plate has been rigid between the sampling areas (C and G respectively) since the Lower Cretaceous.

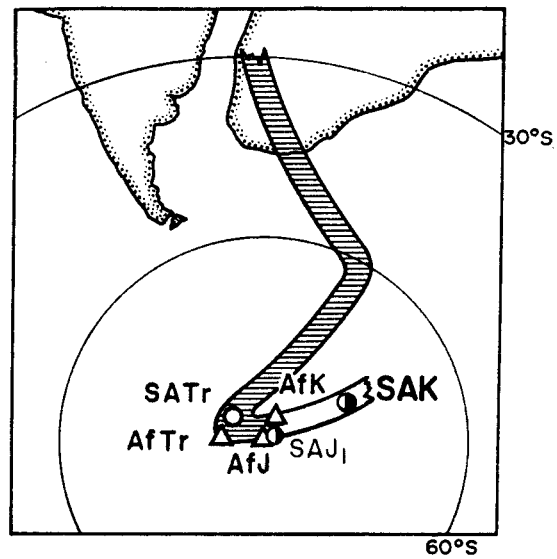


FIG. 5. Late Palaeozoic and Early Mesozoic relative positions of South America and Africa (Valencio & Vilas, 1972). The Middle Jurassic–Lower Cretaceous latitude grid is shown. The common polar wandering curve (full line, stripped) and the Middle Jurassic Lower Cretaceous polar wandering for South America (full line, open) are shown. Palaeomagnetic poles indicated for South America (○) and Africa (△) are either abridged in Table 2 or quoted in Section 4.

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The four Lower Cretaceous palaeomagnetic poles for South America given in Table 2 are reasonably well grouped; the mean of the positions of these four poles (SAK, Table 2, entry 6) defines with increased precision the Lower Cretaceous pole position relative to South America.

Fig. 5 shows the Late Palaeozoic–Early Mesozoic relative positions of South America and Africa (Valencio & Vilas 1972). The Late Palaeozoic–Early Mesozoic polar wandering curve and the Mesozoic mean palaeomagnetic poles for the South American–African block are also shown in Fig. 5. The mean of the palaeomagnetic Triassic and Jurassic poles for Africa (AfTr and AfJ respectively, Valencio & Vilas 1969) overlap with the mean palaeomagnetic poles of the same age (SATr, Valencio & Vilas 1972; and SAJ, Vilas 1974) for South America. This suggests that up to the Middle Jurassic (Bathonian), South America and Africa were joined by their Atlantic margins as indicated in Fig. 5.

Fig. 5 shows that the position of the mean of the Cretaceous palaeomagnetic poles for Africa (AfK, Valencio & Vilas 1969) overlaps with that of the mean of the Jurassic palaeomagnetic poles for that continent (AfJ). That suggest that Africa had a quasi-static interval (Briden 1967) in Jurassic–Cretaceous time. On the other hand in that figure is also shown the discrepancy (15°) between the positions of the mean of the Lower Cretaceous palaeomagnetic poles (SAK) and the Middle Jurassic palaeomagnetic pole (SAJ_1) for South America. This defines a Middle Jurassic–Lower Cretaceous polar shift for South America (white curve in Fig. 5), This polar shift is obviously associated with the fragmentation of the South America–African block and the origin of the South Atlantic. That is consistent with the Upper Valanginian marine transgression recorded in South Africa and Southern South America (Reyment & Tait 1972) and the sea-floor spreading data (Maxwell *et al.* 1970; Larson & Ladd 1973).

Acknowledgments

The author wish to thank the Consejo Nacional de Investigaciones Cientificas y Técnicas and the Universidad de Buenos Aires, Argentina, for the financial support which enabled the work described to be carried on.

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